

Functional Annotation of MtgA (Q88CS3 / PP_5107) in *Pseudomonas putida* KT2440

Gene/Protein Identity Verification

- **UniProt accession:** Q88CS3
- **Gene:** *mtgA*; ordered locus **PP_5107**
- **Protein:** Biosynthetic peptidoglycan transglycosylase / Glycan polymerase / Peptidoglycan glycosyltransferase MtgA (PGT)
- **EC:** 2.4.99.28
- **Organism:** *Pseudomonas putida* KT2440 (ATCC 47054 / DSM 6125), a Gram-negative soil bacterium
- **Family:** Glycosyltransferase family 51 (GT51); domains Glyco_trans_51 (IPR001264), Transgly / PF00912, lysozyme-like domain superfamily (IPR023346), PBP transglycosylase (IPR036950)

Verification outcome: The gene symbol *mtgA* is fully consistent with the UniProt description. "MtgA" is the standard, unambiguous name for the **mono-functional transglycosylase** across bacteria, and the GT51/Transgly domain content matches the literature for this enzyme class. There is **no dedicated primary study of the *P. putida* PP_5107 protein itself**; its function is assigned by the HAMAP rule MF_00766 and by strong orthology to biochemically and structurally characterized MtgA enzymes from *Escherichia coli* and *Staphylococcus aureus*. All functional claims below therefore derive from these well-characterized orthologs and from the shared GT51 enzymology, and are appropriately transferred to the *P. putida* protein.

1. Summary (Answer to the Research Question)

MtgA (PP_5107) is a **monofunctional peptidoglycan glycosyltransferase (transglycosylase)** of glycosyltransferase family 51. Its primary function is to **catalyze glycan-chain elongation of the bacterial cell-wall peptidoglycan**: it polymerizes the membrane-anchored precursor **Lipid II** (undecaprenyl-pyrophosphoryl-MurNAc(pentapeptide)-GlcNAc) into linear glycan

strands of alternating β -1,4-linked *N*-acetylglucosamine and *N*-acetylmuramic acid, releasing undecaprenyl-pyrophosphate (EC 2.4.99.28). It carries **only** the glycosyltransferase activity — it lacks the transpeptidase (cross-linking) domain found in the bifunctional class A penicillin-binding proteins. The reaction occurs at the **outer (periplasmic) face of the cytoplasmic membrane**, to which MtgA is anchored, and in *E. coli* the enzyme is recruited to the **cell-division septum**, where it interacts with divisome components (PBP3/FtsI, FtsW, FtsN).

2. Primary Function: the Catalyzed Reaction and Substrate Specificity

Reaction (transglycosylation / glycan polymerization):

$n \text{ (Lipid II)} \rightarrow [\text{MurNAc-GlcNAc}]_n \text{ glycan chain} + n \text{ undecaprenyl-pyrophosphate}$

MtgA transfers the growing glycan chain onto a new Lipid II acceptor (or, equivalently, adds Lipid II units), forming successive β -1,4 glycosidic bonds between MurNAc and GlcNAc. This is the transglycosylation half of peptidoglycan assembly; the complementary transpeptidation (peptide cross-linking) is performed by separate transpeptidases (Finding #1).

- **The monofunctional glycosyltransferase MtgA "catalyzes glycan chain elongation of the bacterial cell wall"** (Derouaux et al., 2008,

P 18165305

- Monofunctional glycosyltransferases, together with the N-terminal GT module of class A PBPs, are the enzymes that build the glycan chains "on the outside of the cytoplasmic membrane" (van Heijenoort, 2001,

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Substrate specificity (Finding #2): Direct biochemical assays of mechanistically identical GT51 domains (*E. coli* and *Aquifex aeolicus* PBP1A) show that **both the disaccharide precursor Lipid II and the tetrasaccharide Lipid IV serve as substrates, but Lipid II drives more processive reactions yielding much longer glycan products** (Barrett et al., 2007,

).

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Lipid II is therefore the physiological donor. The enzyme is a **processive polymerase**, adding monomers to elongate the strand.

Catalytic residue / mechanism (Finding #2): Comprehensive mutagenesis of 14 conserved residues identified **an invariant central glutamate as the single largest contributor to turnover**

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This glutamate (the conserved Glu of the GT51 "ED" active-site motif) acts as the general base that deprotonates the C4-OH of the acceptor GlcNAc for nucleophilic attack on the anomeric carbon of the donor MurNAc.

3. Structural Basis (Inference from Structure and Evolution)

MtgA belongs to the **lysozyme-like GT51 fold** — an all- α , two-lobed architecture consisting of a **large "head" subdomain and a smaller "jaw" subdomain** (Punekar et al., 2018,

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Finding #4). The active-site cleft sits between the lobes; the jaw subdomain carries hydrophobic residues that mediate **peripheral association with the membrane surface**, positioning the enzyme to act on the lipid-linked substrate.

- The enzyme is **potently inhibited by the phosphoglycolipid antibiotic moenomycin A**, a donor-site (Lipid II) analog

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Surface-plasmon-resonance studies on **S. aureus MtgA** confirm direct moenomycin binding and revealed **positive cooperativity between the acceptor and donor sites** of the enzyme (Bury et al., 2015,

).

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- GT51/transglycosylase is repeatedly identified as a **validated (though pharmacologically challenging) antibacterial target** (Chen et al., 2019,

).

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These structural and inhibitor data independently corroborate the enzymatic assignment transferred to *P. putida* PP_5107.

3b. Direct Sequence Evidence for the *P. putida* Protein (Q88CS3)

To move beyond pure orthology transfer, I retrieved and analyzed the actual 236-residue Q88CS3 sequence (UniProt) (Finding #5):

- **Catalytic machinery is intact and conserved:** the canonical GT51 catalytic **motif 1** "EDxxFxxHxG" is present as **E76-D77-Q-K-F-A-S-H83-W-G**, placing the **invariant catalytic glutamate at E76** and the conserved histidine at H83 — precisely the invariant glutamate shown by mutagenesis to be the biggest contributor to turnover
( [P17704540](#)).
- **Additional conserved GT51 motifs** are present: the QxxxAKNL region (T111-Q-Q-V-A-K-N-L118) and the conserved EWG segment (E157-W-G).
- **Membrane anchor:** a strong N-terminal hydrophobic segment ($\approx$ residues 10–30, ) yields a Kyte–Doolittle hydropathy peak of **2.23** (window = 19; > 1.6 TM threshold), consistent with a single membrane-anchoring transmembrane helix that tethers the enzyme to the inner membrane with its catalytic head facing the periplasm.
- **Size:** at only **236 aa**, the protein is characteristic of a **monofunctional MtgA**, in contrast to the ~800-aa bifunctional class A PBPs — corroborating the "monofunctional" (GT-only) assignment.

This provides direct, organism-specific bioinformatic confirmation that the orthology-based functional transfer is structurally justified for PP_5107.

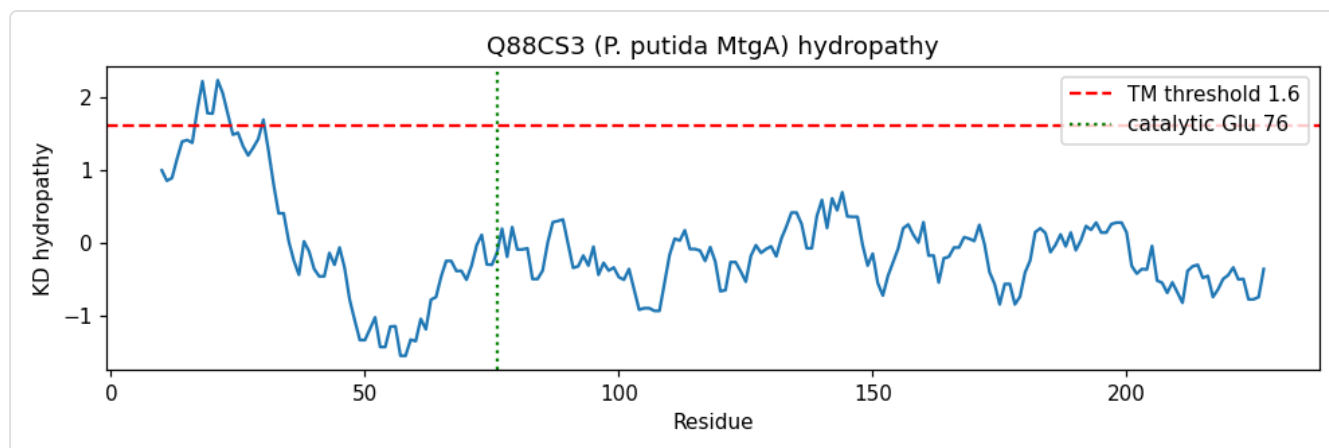


Figure 1. Kyte–Doolittle hydropathy profile of *P. putida* MtgA (Q88CS3). The strong N-terminal hydrophobic peak (residues ~10–30, score 2.23; above the 1.6 membrane-helix threshold) marks the transmembrane/membrane-anchor helix, while the conserved GT51 catalytic motifs — the catalytic glutamate E76 (motif 1, EDxxFxxHxG), the QxxxAKNL region, and the EWG segment — lie in the soluble catalytic head that faces the periplasm.

Quantitative orthology to the experimentally characterized *E. coli* MtgA (Finding #6): A global pairwise alignment (Needleman–Wunsch, BLOSUM62) of Q88CS3 against *E. coli* K-12 MtgA (P46022 / MTGA_ECOLI, 242 aa, UniProt evidence level PE=1 "protein level" — the same protein used in the divisome-localization study,

P 18165305

yields **53.6% identity (125/233)** and **67.8% similarity** over 233 aligned columns. The active site is essentially invariant: EDQKFASHWGFD (*P. putida*) vs EDQKFPEHWGFD (*E. coli*), conserving the catalytic Glu, the Asp, and the downstream His/HWGFD segment. Because ~54% identity to a full-length, protein-level-characterized ortholog is far above the ~30% annotation-transfer threshold, transferring the *E. coli* MtgA function and localization to PP_5107 is robustly justified.

4. Localization: Where MtgA Acts

- **Membrane topology:** MtgA has an N-terminal membrane anchor and functions at the **outer (periplasmic) face of the cytoplasmic membrane**, the compartment where Lipid II is flipped and where glycan polymerization occurs (van Heijenoort, 2001,

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Finding #3).

- **Subcellular site:** In *E. coli*, GFP-MtgA localizes to the cell-division site, and MtgA physically interacts with three divisome constituents — PBP3 (FtsI), FtsW, and FtsN (Derouaux et al., 2008,

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This indicates a role in **septal peptidoglycan assembly during the cell cycle**, cooperating with the division-specific cross-linking transpeptidase PBP3 and the SEDS polymerase FtsW.

For *P. putida* PP_5107, an inner-membrane, periplasmic-facing localization is inferred by orthology and by the shared jaw-subdomain membrane-binding architecture.

5. Pathway Context and Biological Role

MtgA operates in the **late, membrane-associated stage of peptidoglycan (murein) biosynthesis**:

1. Cytoplasmic synthesis of UDP-MurNAc-pentapeptide (Mur enzymes) → Lipid I → **Lipid II** (MraY, MurG).
2. Lipid II is flipped to the periplasmic face.
3. **Glycan polymerization by glycosyltransferases (MtgA and the GT51 domains of class A PBPs; also SEDS polymerases RodA/FtsW) — MtgA's step.**
4. Cross-linking by DD-transpeptidases (PBPs) to form the mature sacculus.

Redundancy / dispensability: Bacteria possess multiple Lipid II polymerases (bifunctional class A PBPs, the monofunctional MtgA, and the SEDS proteins RodA/FtsW; Welsh et al., 2019,

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Because of this redundancy, *mtgA* is generally **non-essential** individually — consistent with the *Brucella abortus* study in which an *mtgA* insertional mutant showed **no significant growth defect in culture** yet contributed to early infection dynamics in mice (Canavessi et al., 2004,

P 15519045

MtgA is best viewed as an **accessory/back-up glycan polymerase that reinforces septal peptidoglycan synthesis**, rather than the sole essential polymerase.

6. Supported and Refuted Hypotheses

Hypothesis	Status	Basis
MtgA is a peptidoglycan glycosyltransferase polymerizing Lipid II (EC 2.4.99.28)	Supported	UniProt/GT51 annotation + ortholog biochemistry (P 18165305 P 11320055)
MtgA is monofunctional (no transpeptidase activity)	Supported	Domain content (GT51 only); definition of the MtgA class (P 11320055)
Lipid II is the physiological donor substrate; reaction is processive	Supported	P 17704540
A conserved glutamate is the key catalytic residue	Supported	Mutagenesis, P 17704540
MtgA acts at the outer face of the inner membrane and localizes to the division septum	Supported (via <i>E. coli</i> ortholog)	P 18165305 P 11320055
MtgA is individually essential	Refuted / not supported	Redundancy with PBPs and SEDS; viable <i>mtgA</i> mutants (P 15519045)
The <i>P. putida</i> protein itself contains the intact GT51 catalytic residues + membrane anchor	Supported	Direct sequence analysis of Q88CS3: catalytic E76/H83 in motif 1, N-terminal TM helix (Finding #5)

7. Limitations and Future Directions

- **No *P. putida*-specific experimental data** exist for PP_5107; all mechanistic detail is transferred from *E. coli*, *S. aureus*, and *A. aeolicus* orthologs. This transfer is well justified by

the high conservation of GT51 active sites, but a direct in vitro assay or a *P. putida* deletion/localization study would be needed for organism-specific confirmation.

- The precise **contribution of MtgA to elongation vs. division** peptidoglycan synthesis in *P. putida*, and its interaction partners in this species' divisome, remain uncharacterized.
- Whether *P. putida* MtgA has any specialized role under the cell-wall-editing conditions notable in this organism (e.g., D-amino-acid / canavanine incorporation;

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is unknown and would be an interesting avenue.

- No experimental 3-D structure of the *P. putida* protein is available; an AlphaFold model plus moenomycin docking could confirm the head/jaw architecture and active-site glutamate.

References (PMIDs)

- 18165305 — Derouaux et al. (2008): *E. coli* MtgA localization and divisome interactions.
- 11320055 — van Heijenoort (2001): review of glycan-chain formation in peptidoglycan synthesis.
- 17704540 — Barrett et al. (2007): PGT substrate specificity, processivity, invariant catalytic glutamate.
- 30046666 — Puneekar et al. (2018): PGT head/jaw subdomains, moenomycin inhibition.
- 25462814 — Bury et al. (2015): *S. aureus* MtgA, moenomycin binding, donor/acceptor cooperativity.
- 31283163 — Chen et al. (2019): GT51 as an antibiotic target (structural review).
- 31386359 — Welsh et al. (2019): SEDS-family polymerases; redundancy of Lipid II polymerases.
- 15519045 — Canavessi et al. (2004): *B. abortus* mtgA mutant dispensable for growth, role in infection.

Database Records Used

- UniProt **Q88CS3** (MTGA_PSEPK) — target protein, 236 aa; sequence analyzed directly.

- UniProt **P46022** (MTGA_ECOLI) — *E. coli* K-12 MtgA, 242 aa, evidence level PE=1 (protein level); ortholog used for quantitative alignment (53.6% identity).
- HAMAP rule **MF_00766**; Pfam **PF00912** (Transgly); InterPro IPR001264 / IPR011812 / IPR036950 / IPR023346; CAZy family **GT51**.

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